How to Push the Project to GitHub
Minor Project 03: Design and Simulation of a 4-Bit Counter

Thipperudra K T | 4SM23EE037 | 24 October 2026

Project Requirements:

1. Objective:

- Upload all project files (code + simulation results) to a GitHub repository and submit the shareable link.

2. Files to Upload:

Folder / file	Content
src/counter_4bit.v	HDL code (Verilog) for the 4-bit counter
tb/tb_counter_4bit.v	Testbench for functional simulation
sim/counter_4bit.vcd	Waveform dump file (open with GTKWave)
sim/waveform_full.png, sim/waveform_zoom.png	Waveform output screenshots
report/Minor_Project_03_4-Bit_Counter_Report.pdf	Project report
README.md, Makefile	Project description and run commands

```
Minor_Project_03_4bit_Counter/
|-- src/      counter_4bit.v
|-- tb/       tb_counter_4bit.v
|-- sim/      counter_4bit.vcd, waveform_full.png, waveform_zoom.png
|-- report/   Minor_Project_03_4-Bit_Counter_Report.pdf
|-- Makefile
|-- README.md
```

3. Waveform Output (stored in the sim folder):

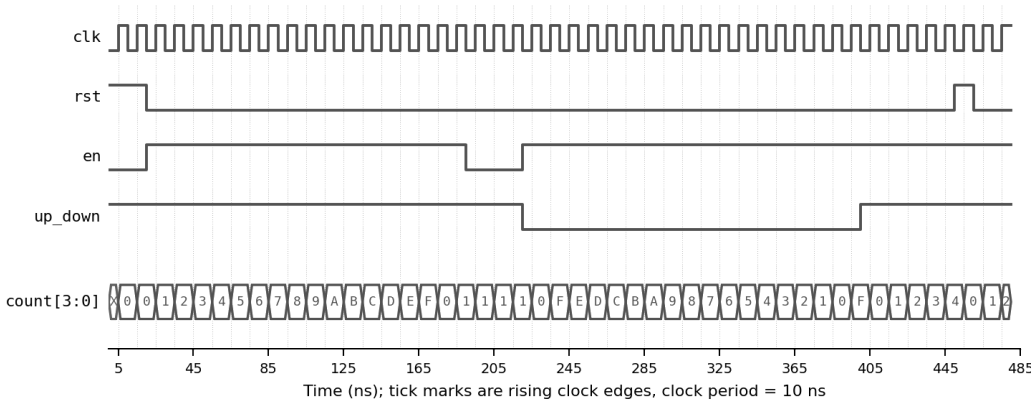


Fig. 1: Complete waveform of count[3:0] with clk, rst, en and up_down (sim/waveform_full.png)

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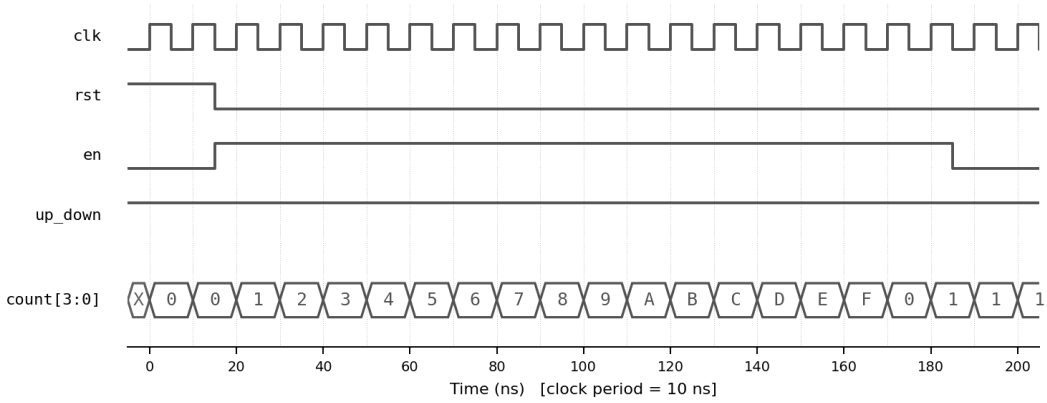


Fig. 2: Zoomed waveform: reset release, count up to F, wrap-around to 0 (sim/waveform_zoom.png)

These images were generated from a cycle-accurate model of the code. Replace them with screenshots from your simulator (Icarus/GTKWave or Vivado) before pushing.

4. Step 1 - Install and Set Up Git:

- Install Git from git-scm.com and check it with:

```
git --version
```

- Tell Git who you are (one time only):

```
git config --global user.name  "Thipperudra K T"
git config --global user.email "your-email@example.com"
```

5. Step 2 - Create the Repository on GitHub:

- Sign in at github.com and click New repository.
- Repository name: Minor_Project_03_4bit_Counter
- Visibility: Public, so the link has no restricted access.
- Leave Add a README unticked (the folder already has one), then click Create repository.

6. Step 3 - Push the Project:

- Open a terminal in the folder that contains Minor_Project_03_4bit_Counter and run:

```
cd Minor_Project_03_4bit_Counter
git init
git add .
git commit -m "Minor Project 03: 4-bit counter (Verilog)"
git branch -M main
git remote add origin https://github.com/<your-username>/Minor_Project_03_4bit_Counter.git
git push -u origin main
```

  - git init - creates a local repository in the folder.
  - git add . - stages all files for the commit.
  - git commit - saves the staged files with a message.
  - git branch -M main - names the branch main.
  - git remote add origin - links the folder to your GitHub repository.
  - git push -u origin main - uploads the commit to GitHub.

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- Replace <your-username> with your GitHub user name. When asked for a password, paste a personal access token (GitHub Settings, Developer settings, Personal access tokens); the account password is not accepted. SSH keys also work.

7. Step 4 - Check the Result:

```
git status
git log --oneline
git remote -v
```

- git status should say the working tree is clean and the branch is up to date with origin/main.
- Open https://github.com/<your-username>/Minor_Project_03_4bit_Counter in a private (incognito) window. If you can see src, tb, sim and report without logging in, the link is accessible.

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8. Alternative - Upload from the Browser:

- Open the new repository, click Add file, then Upload files.
- Drag the src, tb, sim and report folders, README.md and Makefile into the page so the folder structure is kept.
- Enter a commit message such as Add project files and click Commit changes.

9. Updating the Repository Later:

```
git add .
git commit -m "Add simulator waveform screenshots"
git push
```

10. Common Errors and Fixes:

Message	Cause and fix
Authentication failed	Password was used. Use a personal access token as the password, or set up SSH.
remote origin already exists	Run git remote set-url origin <repository-url> and push again.
src refspec main does not match any	No commit yet. Run git add . and git commit first, then git branch -M main.
rejected ... fetch first	The repository already has files. Run git pull --rebase origin main, then git push.
Repository not found	Check the user name and repository name in the URL and that you are signed in.
Files not visible without login	The repository is private. Settings, General, Danger Zone, Change visibility, Make public.

11. Submit the Link:

- Submit the repository link: https://github.com/<your-username>/Minor_Project_03_4bit_Counter
- Individual submission only; make sure the link is accessible (no restricted access) and the file structure is maintained as shown above.

Shareable link: _____

Submission Date: 24/10/2026

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